Package ‘LRg’

June 9, 2026

Title Logistic Regression

Version 0.0.0.9000

Description Logistic regression (one paragraph).

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Encoding UTF-8

Roxygen list(markdown = TRUE)

RoxygenNote 7.3.3

Suggests testthat (>= 3.0.0)

Config/testthat/edition 3

LazyData true

Imports maxLik

Contents

auto_factor *Automatic recoding into factors*

Description

Automatic recoding into factors

Usage

```
auto_factor(formula, data, max_levels = 10)
```

Arguments

`formula`	model expression
`data`	data frame

Value

recoded data frame

btls *Backtracking line search*

Description

Backtracking line search

Usage

```
btls(x, X, Z, dx, u, alpha = 1, rho = 0.5, c = 1e-04, maxit = 20)
```

Arguments

`x`	parameter vector
`X`	design matrix
`Z`	response dummies
`dx`	search direction
`u`	score vector
`alpha`	step size
`rho`	contraction factor
`c`	Armijo parameter
`maxit`	maximum number of iterations

Value

new parameter vector

`btls_stn`	*Backtracking line search*

Description

Backtracking line search

Usage

```
btls_stn(x, X, Z, q, dx, u, alpha = 1, rho = 0.5, c = 1e-04, maxit = 20)
```

Arguments

`x`	parameter vector
`X`	design matrix
`Z`	response dummies
`dx`	search direction
`u`	score vector
`alpha`	step size
`rho`	contraction factor
`c`	Armijo parameter
`maxit`	maximum number of iterations

Value

new parameter vector

`cum_contr`	*Cumulative binary recoding*

Description

Cumulative binary recoding

Usage

```
cum_contr(x)
```

Arguments

`x`	ordered factor

Value

cumulative contrast

`eyedisease` — *Empirical dataset*

Description

The data are 720 observations on 6 variables.

Usage

```
eyedisease
```

Format

A data frame with 720 rows and 6 columns

diab diabetes

gh glycosylated haemoglobin level

dbp diastolic blood pressure

prot proteinuria

merl maximum severity of retinopathy in the left and right eye

Source

From the Wisconsin Epidemiological Study of Diabetic Retinopathy (Klein et al., 1984). From R package ordinalgmifs (Archer et al., 2014).

`jacob` — *Jacobian*

Description

Jacobian

Usage

```
jacob(alpha, gamma, Lambda, Phi)
```

Arguments

`alpha`	vector of intercepts
`gamma`	vector of regressor effects
`Lambda`	matrix of regressor effects
`Phi`	matrix of weights

Value

Jacobian matrix

jacob_stn — *Jacobian*

Description

Jacobian

Usage

```
jacob_stn(Phi, Gamma)
```

Arguments

`Phi`	matrix
`Gamma`	matrix

Value

Jacobian

logLik — *The log-likelihood function*

Description

The log-likelihood function

Usage

```
logLik(x, X, Z)
```

Arguments

`x`	parameter vector
`X`	desgin matrix
`Z`	response dummies

Value

log-likelihood value

`logLik_stn` *The log-likelihood function*

Description

The log-likelihood function

Usage

```
logLik_stn(x, X, Z, q)
```

Arguments

`x`	parameter vector
`X`	desgin matrix
`Z`	response dummies

Value

log-likelihood value

`lrt` *Likelihood ratio test*

Description

Likelihood ratio test

Usage

```
lrt(obj1, obj2)
```

Arguments

`obj1`	fit results model 1
`obj2`	fit results model 2

Value

test statistic, df, and p-value

mlr *Multinomial logistic regression*

Description

Multinomial logistic regression

Usage

```
mlr(formula, data, maxit = 100)
```

Arguments

`formula`	model expression
`data`	data frame
`maxit`	maximum number of iterations

Value

A list of

`coefficients`	Parameter estimates
`fitted.values`	The model implied probabilities
`residuals`	Dummy responses minus fitted values
`pred.cat`	The model implied categories

olr *Ordinal logistic regression*

Description

Ordinal logistic regression

Usage

```
olr(formula, data, maxit = 100)
```

Arguments

`formula`	model expression
`data`	data frame
`maxit`	maximum number of iterations

Value

A list of

`coefficients`	Parameter estimates
`fitted.values`	The model implied probabilities
`residuals`	Dummy responses minus fitted values
`pred.cat`	The model implied categories

pac — *Parallel adjacent-categories logit model*

Description

Parallel adjacent-categories logit model

Usage

```
pac(formula, data, maxit = 100)
```

Arguments

`formula`	model expression
`data`	data frame
`maxit`	maximum number of iterations

Value

A list of

`coefficients`	Parameter estimates
`fitted.values`	The model implied probabilities
`residuals`	Dummy responses minus fitted values
`pred.cat`	The model implied categories

sim_data — *Example dataset*

Description

A data set containing a single ordinal response variable and four continuous regressor variables. The response variable has four levels.

Usage

```
sim_data
```

Format

A data frame with 300 rows and 5 columns

y Response

X1 to X4 Regressor variables

Source

Simulated data

`solve_safe` *Solving Idx=u*

Description

Solving Idx=u

Usage

```
solve_safe(I, u, lambda = 1e-08)
```

Arguments

`I`	information matrix or Hessian
`u`	score or gradient
`lambda`	ridge penalty parameter

Value

ridge information matrix or Hessian

`stn` *Stereotype nominal regression*

Description

Stereotype nominal regression

Usage

```
stn(formula, q = NULL, data, stv = NULL, csv = 0.001, tol = 1e-08, maxit = 100)
```

Arguments

`formula`	model expression
`q`	rank
`data`	data frame
`stv`	vector of starting values
`csv`	common starting value
`tol`	convergence tolerance
`maxit`	maximum number of iterations

Value

A list of

`alpha`, `Beta`, `Gamma`, `Phi`	Parameter estimates
`loglik`	Value of the log-likelihood function
`npars`	Number of estimated parameters
`fitted.values`	Model implied probabilities
`residuals`	Dummy responses minus fitted values
`pred.cat`	Model implied categories
`obs.cat`	Observed categories

`sto` *Stereotype ordered regression*

Description

Stereotype ordered regression

Usage

```
sto(formula, q = NULL, data, tol = 1e-08, maxit = 100)
```

Arguments

`formula`	model expression
`q`	rank
`data`	data frame
`tol`	convergence tolerance
`maxit`	maximum number of iterations

Value

A list of

`alpha`, `Beta`, `Gamma`, `Phi`	Parameter estimates
`loglik`	Value of the log-likelihood function
`npars`	Number of estimated parameters
`fitted.values`	Model implied probabilities
`residuals`	Dummy responses minus fitted values
`pred.cat`	Model implied categories
`obs.cat`	Observed categories

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